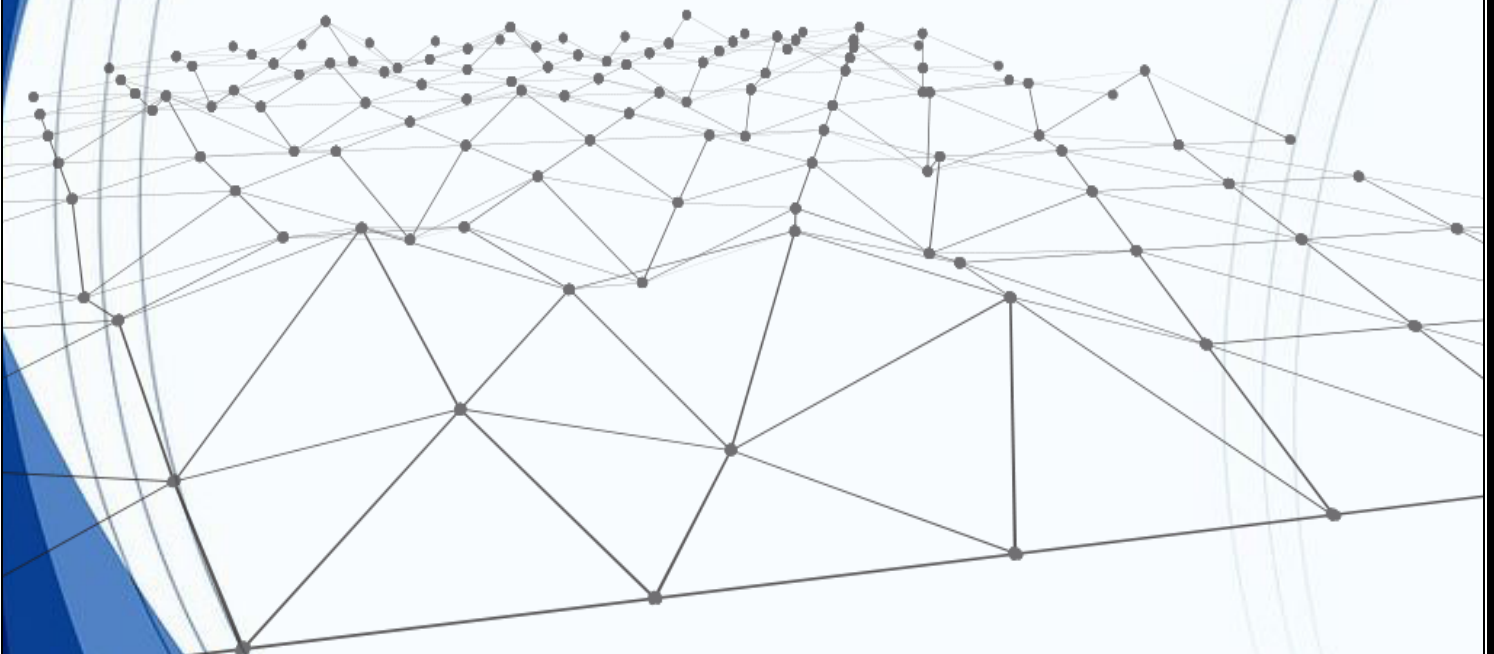




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August, 2026

New York City Motor Collision EDA



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1. Executive Summary

Project scope: Analysis of New York City motor vehicle collision data (2012–2026) covering over 2.26 million crash records.

```
df = pd.read_csv('/kaggle/input/datasets/trietes/motor-vehicle-collisions-crashes/Motor Vehicle Collisions - 2012-2026.csv')
df.head()
```

	CRASH DATE	CRASH TIME	BOROUGH	ZIP CODE	LATITUDE	LONGITUDE	LOCATION	ON STREET NAME	CROSS STREET NAME	OFF STREET NAME
0	09/11/2021	2:39	NaN	NaN	NaN	NaN	NaN	WHITESTONE EXPRESSWAY	20 AVENUE	NaN
1	03/26/2022	11:45	NaN	NaN	NaN	NaN	NaN	QUEENSBORO BRIDGE UPPER	NaN	NaN
2	11/01/2023	1:29	BROOKLYN	11230	40.62179	-73.970024	(40.62179, -73.970024)	OCEAN PARKWAY	AVENUE K	NaN
3	06/29/2022	6:55	NaN	NaN	NaN	NaN	NaN	THROGS NECK BRIDGE	NaN	NaN
4	09/21/2022	13:21	NaN	NaN	NaN	NaN	NaN	BROOKLYN BRIDGE	NaN	NaN

5 rows x 29 columns

Core Takeaways: Overall crash severity rates have steadily increased across all boroughs over time, with distinct victim impact profiles by geographic region (e.g., pedestrian/cyclist prevalence in Manhattan vs. motorist dominance in outer boroughs).

2. Data Quality & Preprocessing

Dataset Overview: Evaluation of dataset dimensions (2,269,187 rows, 29 columns) and initial data types.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2269187 entries, 0 to 2269186
Data columns (total 42 columns):
#   Column                                     Dtype
---  -
0   CRASH DATE                               object
1   CRASH TIME                               object
2   BOROUGH                                  object
3   ZIP CODE                                 object
4   LATITUDE                                float64
5   LONGITUDE                               float64
6   LOCATION                                object
7   ON STREET NAME                          object
8   CROSS STREET NAME                       object
9   OFF STREET NAME                         object
10  NUMBER OF PERSONS INJURED                float64
11  NUMBER OF PERSONS KILLED                 float64
12  NUMBER OF PEDESTRIANS INJURED            int64
13  NUMBER OF PEDESTRIANS KILLED             int64
14  NUMBER OF CYCLIST INJURED                int64
15  NUMBER OF CYCLIST KILLED                 int64
16  NUMBER OF MOTORIST INJURED               int64
17  NUMBER OF MOTORIST KILLED                int64
18  CONTRIBUTING FACTOR VEHICLE 1             object
19  CONTRIBUTING FACTOR VEHICLE 2             object
20  CONTRIBUTING FACTOR VEHICLE 3             object
21  CONTRIBUTING FACTOR VEHICLE 4             object
22  CONTRIBUTING FACTOR VEHICLE 5             object
23  COLLISION_ID                             int64
24  VEHICLE TYPE CODE 1                      object
25  VEHICLE TYPE CODE 2                      object
26  VEHICLE TYPE CODE 3                      object
27  VEHICLE TYPE CODE 4                      object
28  VEHICLE TYPE CODE 5                      object
29  DATETIME                                 datetime64[ns]
30  YEAR                                     int32
31  MONTH                                    int32
32  DAY                                      int32
33  HOUR                                     int32
34  DAY_OF_WEEK                             int32
35  IS_WEEKEND                              int64
36  HOUR_SIN                                float64
37  HOUR_COS                                float64
38  MONTH_SIN                               float64
39  MONTH_COS                               float64
40  DAY_SIN                                 float64
41  DAY_COS                                 float64
dtypes: datetime64[ns](1), float64(10), int32(5), int64(8), object(18)
memory usage: 683.8+ MB
None
(2269187, 42)
```

```
Index(['CRASH DATE', 'CRASH TIME', 'BOROUGH', 'ZIP CODE', 'LATITUDE',
      'LONGITUDE', 'LOCATION', 'ON STREET NAME', 'CROSS STREET NAME',
      'OFF STREET NAME', 'NUMBER OF PERSONS INJURED',
      'NUMBER OF PERSONS KILLED', 'NUMBER OF PEDESTRIANS INJURED',
      'NUMBER OF PEDESTRIANS KILLED', 'NUMBER OF CYCLIST INJURED',
      'NUMBER OF CYCLIST KILLED', 'NUMBER OF MOTORIST INJURED',
      'NUMBER OF MOTORIST KILLED', 'CONTRIBUTING FACTOR VEHICLE 1',
      'CONTRIBUTING FACTOR VEHICLE 2', 'CONTRIBUTING FACTOR VEHICLE 3',
      'CONTRIBUTING FACTOR VEHICLE 4', 'CONTRIBUTING FACTOR VEHICLE 5',
      'COLLISION_ID', 'VEHICLE TYPE CODE 1', 'VEHICLE TYPE CODE 2',
      'VEHICLE TYPE CODE 3', 'VEHICLE TYPE CODE 4', 'VEHICLE TYPE CODE 5'],
      dtype='object')
```

Missing Value Analysis:

- Imputation of minor missing values in injury/fatality counts with baseline zeros.

```
df['NUMBER OF PERSONS INJURED'] = df['NUMBER OF PERSONS INJURED'].fillna(0)
df['NUMBER OF PERSONS KILLED'] = df['NUMBER OF PERSONS KILLED'].fillna(0)
```

- Handling coordinate anomalies, including filtering out 7,591 invalid (0.0, 0.0) latitude/longitude records.

```
missing_boroughs_mask = df['BOROUGH'].isna() & df['LATITUDE'].notna() & df['LONGITUDE'].notna()
valid_coords_mask = (df['LATITUDE'] != 0) & (df['LONGITUDE'] != 0)

needs_filling_borough = df[missing_boroughs_mask & valid_coords_mask].copy()
len(needs_filling_borough)
```

487174

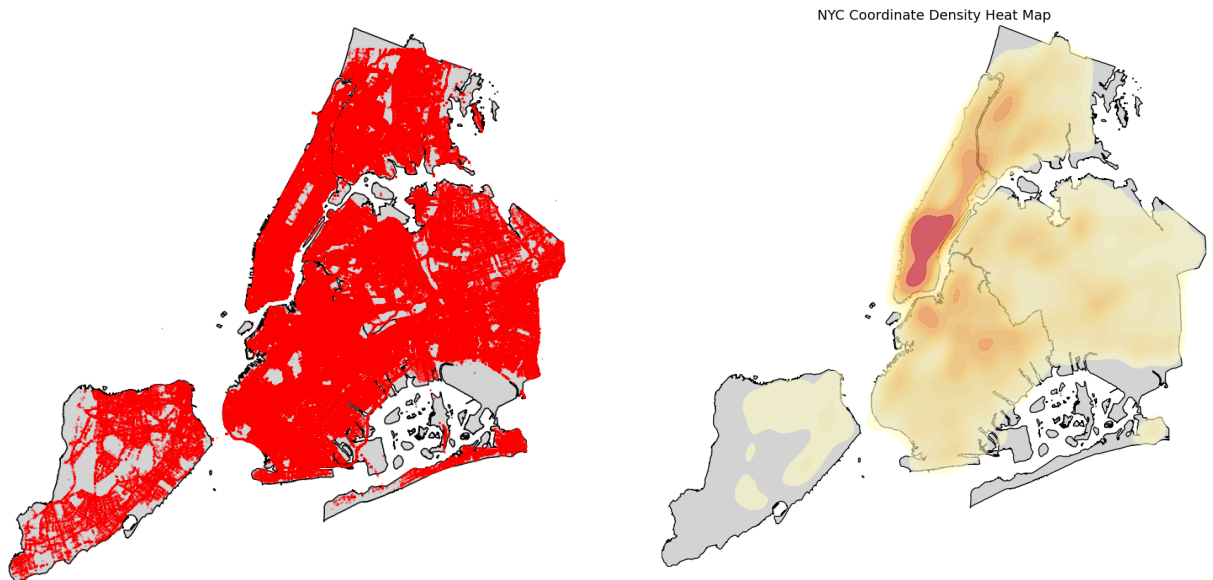
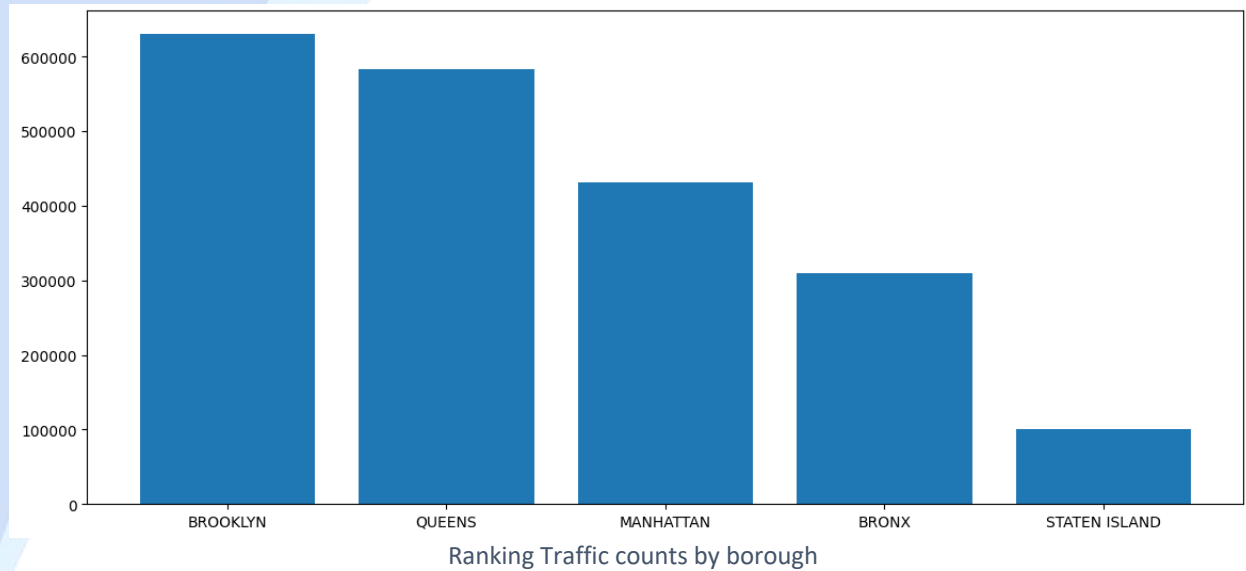
- Geospatial Imputation: Spatial join (EPSG:4326) using official NYC GeoJSON borough boundaries to recover missing borough metadata for over 487,000 records.

DATASETS

- ▶  Motor Vehicle Collisions -
- ▶  NYC Borough GeoJSON

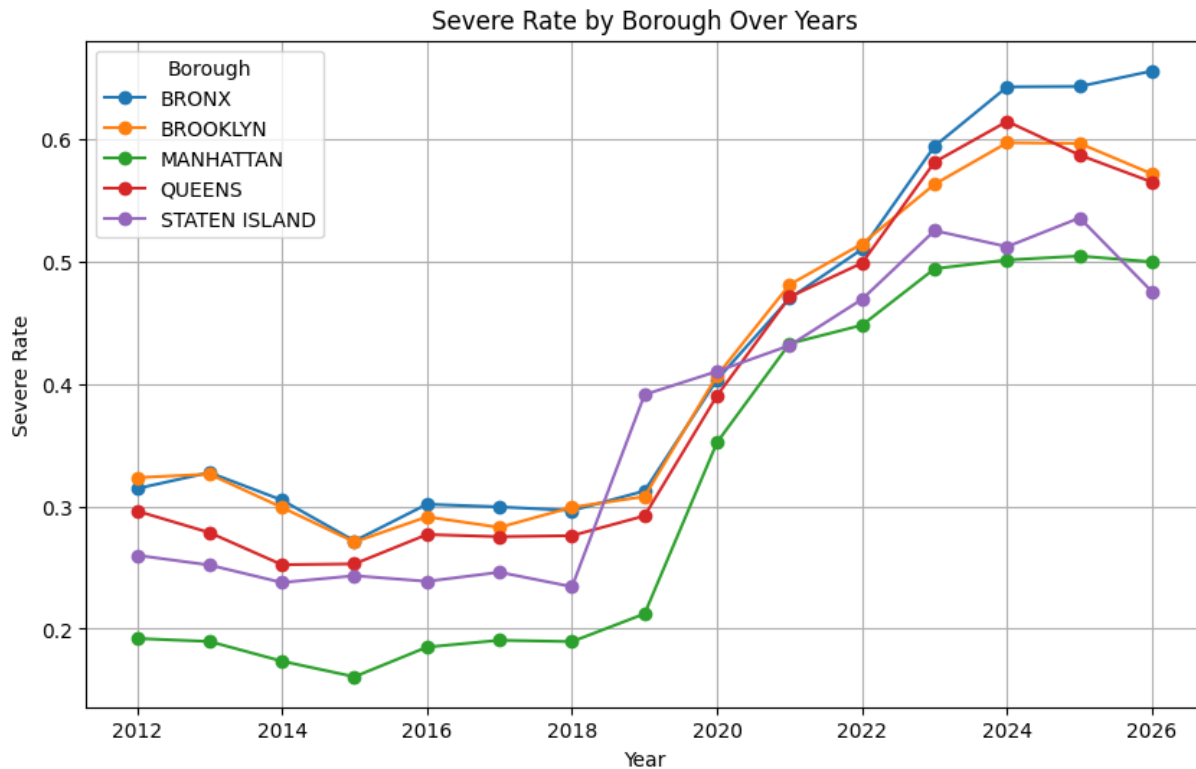
3. Geographic Distribution & Severity Analysis

Volume by Borough: Ranking total collision counts across Brooklyn, Queens, Manhattan, Bronx, and Staten Island.



Crash Severity Trends:

- Longitudinal tracking of injury/mortality rate per crash across boroughs from 2012 through 2026.



- Anomaly cases: Bronx exhibits higher overall severity rates despite lower total collision volume compared to Brooklyn.

4. Victim Demographics & Impact Profile:

Injury Proportion Breakdown: Multi-class proportion analysis of total injuries categorized by Pedestrians, Cyclists, and Motorists per borough:

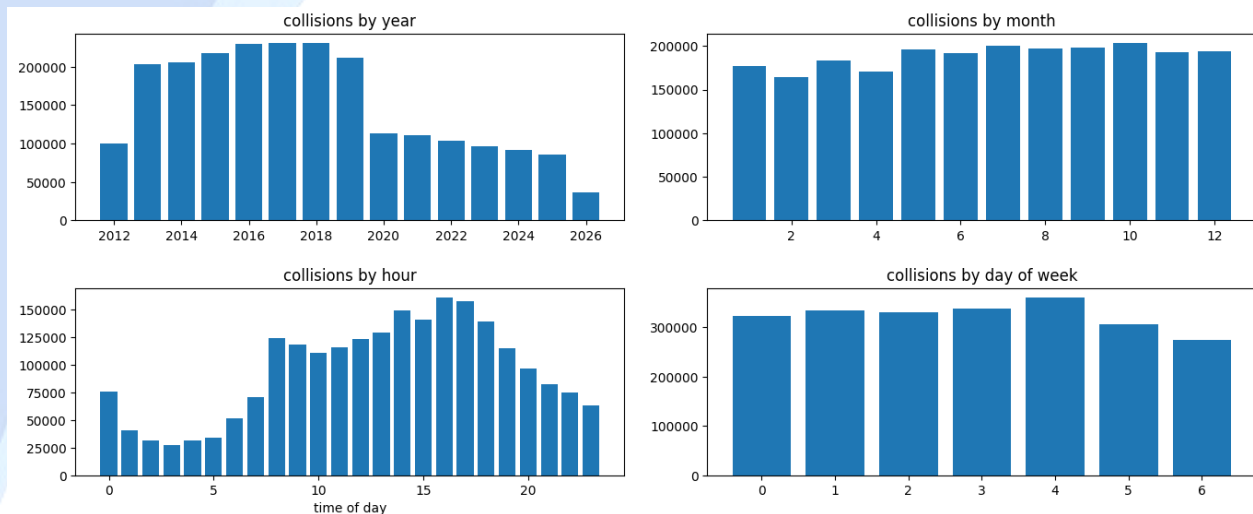
```
(df.groupby('BOROUGH')[['NUMBER OF PEDESTRIANS INJURED', 'NUMBER OF CYCLIST INJURED', 'NUMBER OF MOTORIST INJURED']  
.sum().apply(lambda x: x / x.sum(), axis=1))
```

	NUMBER OF PEDESTRIANS INJURED	NUMBER OF CYCLIST INJURED	NUMBER OF MOTORIST INJURED
BOROUGH			
BRONX	0.188126	0.058831	0.753043
BROOKLYN	0.193923	0.107864	0.698213
MANHATTAN	0.277791	0.184330	0.537879
QUEENS	0.158764	0.061766	0.779470
STATEN ISLAND	0.142580	0.030130	0.827290

- High-density pedestrian/cyclist vulnerability in Manhattan.
- Motorist-dominated injury outcomes in Staten Island, Queens, and the Bronx.

5. Chronological Factor in Collisions:

Key takeaway: There is no correlation between collisions and months/day of week, while collisions by year sees a decline (except incomplete 2026 and 2012) and collisions by hour is at its lowest past midnight and before morning commute hours.



6. Recommendations & Policy Insights

- Borough-specific traffic safety interventions based on victim profile distribution.
- Data collection improvements for spatial recording and street-level metadata.